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Niall L. Williams

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EDUCATION

University of Maryland, College Park, MD, USA — Aug 2019 - Dec 2024
Ph.D. in Computer Science (Dec. 2024)
M.S. in Computer Science (Dec. 2023)
– Research interests: Virtual/Augmented Reality, Human Perception & Navigation, Computer Graphics, Robotics
– Dissertation Title: *Computational Methods for Natural Walking in Virtual Reality*
– Advisors: Dr. Dinesh Manocha, Dr. Aniket Bera, Dr. Ming C. Lin

Davidson College, NC, USA — Aug 2015 - May 2019
B.S. with High Honors in Computer Science
– Thesis Title: Estimation and Comparison of Rotation Gain Thresholds for Redirected Walking
– Advisor: Dr. Tabitha C. Peck

PROFESSIONAL EXPERIENCE (HIGHER EDUCATION)

Barnard College, Columbia University — New York, NY USA
Tenure Track Assistant Professor — *July 2026 - Present*

University of Zaragoza — Zaragoza, Spain
Postdoctoral Researcher (Advisor: **Prof. Ana Serrano)** — *Feb 2026 - June 2026*

New York University Tandon School of Engineering — New York, NY USA
Faculty Fellow (Advisor: **Prof. Qi Sun)** — *Sept 2024 - Sept 2025*

Davidson College — Davidson, NC USA
Research Assistant (Advisor: **Prof. Tabitha C. Peck**) — *May 2018 - Aug 2019*

PROFESSIONAL EXPERIENCE (INDUSTRY)

Human Performance and Experience team, NVIDIA Research — Santa Clara, CA USA
Research Intern (Managers: **Dr. Ruth Rosenholtz, Dr. Jae-Hyun Jung**) — *Jan 2024 - Aug 2024*

Applied Perception Science Team, Meta Reality Labs — Redmond, WA USA
Research Scientist Intern (Managers: **Dr. Ian Erkelens, Dr. Phillip Guan**) — *May 2022 - Aug 2022*

GRANTS & GIFTS

Awarded

1. **University of Central Florida (NSF-supported)**, *"Interaction Effects and Demographic Generalization of VR Display Power-Saving Techniques,"* Virtual Experience Research Accelerator (VERA) Use Grant, Sept 2026 - Dec 2026, Principal Investigator, $3,000

AWARDS & HONORS

Outstanding Reviewer (IEEE ISMAR 2026)	Oct 2026
Nominated for Uni. of Maryland Larry S. Davis Doctoral Dissertation Award (dept. award)	Sept 2024
New York University Tandon CSE Faculty Fellow	March 2024
University of Minnesota President's Postdoctoral Fellowship (declined)	March 2024
Bucknell University Aspiring Primarily Undergraduate Institution Faculty Workshop	Sept 2023
Link Foundation Modeling, Simulation, & Training Fellowship ($34,000)	Aug 2022
Best Paper Honorable Mention (IEEE VR 2022)	March 2022
Meta PhD Research Fellowship Finalist (top 6% out of 2,300+ applicants)	Feb 2022
Best Paper Honorable Mention (IEEE ISMAR 2021)	Oct 2021
Best Paper Honorable Mention (IEEE VR 2021)	March 2021

Dean's Fellowship, University of Maryland ($5,000)	2019, 2020
Senior Computer Science Award, Davidson College	May 2019
Nominated for CRA Outstanding Undergraduate Researcher Award	Oct 2018
Davidson College ASA DataFest Hackathon "Best Use of External Data" Award	April 2017

PUBLICATIONS & INVITED TALKS

Computer science uses competitive conferences (10-30% accepted) as the main publication venue. The top venues in virtual reality (IEEE VR and IEEE ISMAR) offer a track to submit work to be accepted to the conference as well as accepted as part of a special edition of the IEEE Transactions on Visualization and Computer Graphics journal. A full list of my publications can be found on my Google Scholar profile.

* denotes equal contribution. U denotes undergraduate mentee. G denotes graduate mentee.

Journal Papers

[J1] H Kam G, **NL Williams**, J Lubin, JR Bergen, M Sizintsev, Q Sun, C Groth. Attention Drives Cybersickness in Augmented Reality. *IEEE Transactions on Visualization and Computer Graphics*, 2026 (Proc. IEEE ISMAR 2026)

[J2] **NL Williams**, LC Stevens U, A Bera, D Manocha. Sensitivity to Redirected Walking Considering Gaze, Posture, and Luminance. *IEEE Transactions on Visualization and Computer Graphics*, 2025 (Proc. IEEE VR 2025) (17.3% acceptance rate) [link]

[J3] MR Saeedpour-Parizi, **NL Williams**, T Wong, P Guan, D Manocha, IM Erkelens. Perceptual Thresholds for Radial Optic Flow Distortion in Near-Eye Stereoscopic Displays. *IEEE Transactions on Visualization and Computer Graphics*, 2024 (Proc. IEEE VR 2024) (12.6% acceptance rate) [link]

[J4] **NL Williams**, A Bera, D Manocha. Redirected Walking in Static and Dynamic Scenes Using Visibility Polygons. *IEEE Transactions on Visualization and Computer Graphics*, 2021 (Proc. IEEE ISMAR 2021) (19.7% acceptance rate) **[Best paper honorable mention]** [link]

[J5] **NL Williams**, A Bera, D Manocha. ARC: Alignment-based Redirection Controller for Redirected Walking in Complex Environments. *IEEE Transactions on Visualization and Computer Graphics*, 2021 (Proc. IEEE VR 2021) (15.5% acceptance rate) **[Best paper honorable mention]** [link]

[J6] **NL Williams** and TC Peck. Estimation of Rotation Gain Thresholds Considering FOV, Gender, and Distractors. *IEEE Transactions on Visualization and Computer Graphics*, 2019 (Proc. IEEE ISMAR 2019) (8.6% acceptance rate) [link]

Conference Papers

[C1] J Kang G, M Silva, P Sangkloy, K Chen, **NL Williams**, Q Sun. GeneVA: A Dataset of Human Annotations for Generative Text to Video Artifacts. *IEEE/CVF Winter Conference on Applications of Computer Vision*, 2026 (32.6% acceptance rate) [link]

[C2] J Kang G, B Duinkharjav, **NL Williams**, Q Sun. Performance Analysis of Catch-Up Eye Movements in Visual Tracking. *Proceedings of the SIGGRAPH Asia 2025 Conference Papers*, 2025 (27.2% acceptance rate) [link]

[C3] **NL Williams***, N Rewkowski*, J Li, MC Lin. A Framework for Active Haptic Guidance Using Robotic Haptic Proxies. *IEEE International Conference on Robotics and Automation*, 2023 (43.04% acceptance rate) [link]

[C4] **NL Williams**, A Bera, D Manocha. ENI: Quantifying Environment Compatibility for Natural Walking in Virtual Reality. *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2022 (21.5% acceptance rate) **[Best paper honorable mention]** [link]

[C5] JK Terry, B Black, N Grammel, M Jayakumar, A Hari, R Sullivan, LS Santos, C Dieffendahl, C Horsch, R Perez-Vicente, **NL Williams**, Y Lokesh, P Ravi. PettingZoo: Gym for Multi-Agent Reinforcement Learning. *Neural Information Processing Systems (NeurIPS)*, 2021 (26% acceptance rate) [link]

[C6] U Bhattacharya, N Rewkowski, P Guhan, **NL Williams**, T Mittal, A Bera, D Manocha. Generating Emotive Gaits for Virtual Agents Using Affect-Based Autoregression. *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2020 (22.8% acceptance rate) [link]

Workshop Papers and Posters

[P1] **NL Williams**, A Evdokimov, B Duinkharjav, A Patney, Q Sun, J Jung, R Rosenholtz. Detection of artifacts in clean and corrupted video pairs is influenced by artifact type and presentation modality. *Vision Sciences Society*, 2025

[P2] A Gao, X Wang, G Lee, W Chambers, **NL Williams**, Y Qiao, S Xu, MC Lin. Event-Driven Lighting for Immersive Attention Guidance. *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2025

[P3] R Rosenholtz, **NL Williams**. Who you lookin' at? Perception of gaze direction in group settings depends on naturalness of gaze behavior and clutter. *Vision Sciences Society*, 2024

[P4] **NL Williams**, A Bera, D Manocha. Redirection Using Alignment. *IEEE VR Locomotion Workshop*, 2021

[P5] K Qi, D Borland, E Jackson, **NL Williams**, J Minogue, and TC Peck. The impact of haptic and visual feedback on teaching. *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2020

[P6] K Qi, D Borland, **NL Williams**, E Jackson, J Minogue, and TC Peck. Augmenting Physics Education with Haptic and Visual Feedback. *IEEE VR 2020 Fifth Workshop on K-12+ Embodied Learning through Virtual & Augmented Reality (KELVAR)*, 2020

[P7] J Minogue, D Borland, TC Peck, E Jackson, K Qi, and **NL Williams**. Tracing the development of a haptically-enabled science simulation (HESS) for buoyancy. *NARST Annual International Conference*, 2020

[P8] **N Williams** and TC Peck. Estimation of rotation gain thresholds for redirected walking considering FOV and gender. *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, 2019

Invited Talks

[T1] Detection of Video Artifacts in Natural Scenes Depends on Peripheral Visibility,
Google XR Seminar Series, Google, 2025.

[T2] Perception of Motion Artifacts in Near-Eye Varifocal Displays,
UMD Computer Vision Seminar, University of Maryland, 2024. [link]

[T3] Methods for Natural Walking in Virtual Reality,
Research Seminar - MS in Robotics, Graphics and Computer Vision, Universidad de Zaragoza, 2024. [link]

[T4] ARC: Alignment-based Redirection Controller for Redirected Walking in Complex Environments,
SIGGRAPH 2021 TVCG Session on VR, SIGGRAPH 2021. [link]

[T5] Measuring Perceptual Limits of Redirected Walking in Virtual Reality,
Davidson College Coffee Talk, Davidson College, NC, 2018.

TEACHING EXPERIENCE

Computer Science Instructor of Record — Sept 2026 - Present
Barnard College — *New York, NY*

- COMS BC3122: Data Visualization — Fall 2026

Computer Science Instructor of Record — Sept 2024 - Sept 2025
New York University — *New York, NY*

- CS-UY 1114 D: Intro to Programming & Problem Solving — Spring 2025
- CS-GY 6313 B: Information Visualization — Fall 2024

Computer Science Teaching Assistant — Aug 2019 - May 2024
University of Maryland, College Park — *College Park, MD*

- CMSC838C: Advances in XR — Spring 2024
- CMSC425: Game Programming — Fall 2023
- CMSC838C: Advances in XR — Spring 2023
- CMSC838C: Advances in XR — Spring 2022
- CMSC420: Advanced Data Structures — Fall 2021

- CMSC423: Bioinformatic Algorithms, Databases, and Tools — Spring 2021
- CMSC420: Advanced Data Structures — Fall 2020
- CMSC425: Game Programming — Spring 2020
- CMSC420: Advanced Data Structures — Fall 2019

Head Computer Science Teaching Assistant Jan 2019 - May 2019
Davidson College Mathematics & Computer Science Department *Davidson, NC*

Computer Science Tutor Aug 2018 - May 2019
Davidson College Center for Teaching & Learning *Davidson, NC*

- **Courses tutored:** CSC121: Programming and Problem Solving, Discrete Structures, CSC221: Data Structures, CSC250: Computer Organization, CSC209: Bioinformatics Programming.

Computer Science Grader Aug 2017 - Dec 2018
Davidson College Mathematics & Computer Science Department *Davidson, NC*

MENTORING EXPERIENCE

Graduate Students

2024 - Present **Jenna Kang** (PhD), Collaborating on a project that studies people's ability to accurately track an on-screen moving target as a function of target visibility and speed.

2024 - Present **Henry Kam** (PhD), Collaborating on a project studying user's sensitivity to visual motion of virtual objects during locomotion in augmented reality and how erroneous virtual object motion contributes to feelings of motion sickness.

2025 - 2026 **Constantinos Vafias** (MS), Mentored on a project about per-mesh mixed level of detail geometry representation and rendering.

Undergraduate Students

2024 - 2025 **Tolya Evdokimov**: Collaborated on a project that studies human gaze behavior during video quality judgement tasks, to better understand how different video-viewing interfaces influence a person's ability to detect visual artifacts in videos. *Next: Ph.D. student at NYU.*

2021 - 2024 **Logan Stevens**: Collaborated on a project studying correlations between users' eye movements and injected visual motion gains in virtual reality under different brightness conditions. Mentored on a project studying users' sensitivity to visual motion during eye vergence movements in virtual reality. *Next: Ph.D. student at the University of Maryland, College Park [NSF Graduate Research Fellowship (GRFP) recipient]*

2023 - 2024 **Benjamin Margolin**: Mentored on a project focused on developing an algorithm to procedurally generate a virtual environment that allows natural walking for long distances in the user's physical environment. *Next: M.S. student at the University of Maryland, College Park.*

2023 - 2024 **Daniel Lopez**: Mentored on a project that studied the viability of using dynamic, interactive agents in virtual environments to distract the user from injected visual motion gains during locomotion. *Next: Ph.D. student at University of California, Santa Barbara.*

2022 - 2023 **Jason Alexander Fotso-Puepi**, Mentored on a project that studied methods for procedurally generating environment layouts to be used in a large-scale benchmarking test for different virtual reality locomotion interfaces. *Next: Ph.D. student at Purdue University.*

High School Students

Summer 2025 **Aneera Shaikh**, Mentored on a robot motion planning project.

PROFESSIONAL SERVICE & COMMUNITY INVOLVEMENT

Committee Member

ACM SIGGRAPH North America History and Archive Committee	2023 - Present
IEEE VR International Program Committee (papers track)	2027
ACM SIGGRAPH Asia Technical Communications & Posters Committee	2026

	IEEE ISMAR International Program Committee (posters track)	2026
	International Conference on Extended Reality International Program Committee (papers track)	2026
	ACM SIGGRAPH North America "From Human Vision to Machine Vision" Workshop Organizer	2026
	ACM SIGGRAPH North America Posters Juror	2026
	EuroXR International Conference Scientific Program Committee	2026
	IEEE VR Future Faculty Forum Chair	2026
	IEEE VR International Program Committee (posters track)	2026
	IEEE VR International Program Committee (papers track)	2026
	ACM SIGGRAPH North America Posters Juror	2025
	ACM VRST International Program Committee (papers track)	2025
	IEEE ISMAR International Program Committee (papers track)	2025
	IEEE VR Future Faculty Forum Chair	2025
	IEEE ISMAR Future Faculty Forum Chair	2024
	IEEE VR Future Faculty Forum Chair	2024
	IEEE ISMAR Future Faculty Forum Chair	2023
	ACM SIGGRAPH Research Career Development Committee	2021 - 2023
Peer Reviewing	*IEEE Conference on Virtual Reality and 3D User Interfaces (VR)* (2020 - present)	34 reviews
	IEEE Int. Symp. Mixed & Aug. Reality (ISMAR) (2021 - present)	28 reviews
	IEEE Transactions on Visualization & Computer Graphics (TVCG) (2021 - present)	18 reviews
	ACM SIGGRAPH North America & Asia (2022 - present)	5 reviews
	ACM Virtual Reality Software and Technology (VRST) (2023)	2 reviews
	ACM Transactions on Applied Perception (TAP) (2024 - present)	2 reviews
	ACM Transactions on Graphics (TOG) (2024)	1 review
	European Association for Computer Graphics EUROGRAPHICS (2024)	1 review
	PRESENCE: Virtual and Augmented Reality (2024)	1 review
	IEEE Transactions on Games (2022)	1 review
	ACM Conference on Human Factors in Computing Systems (CHI) (2025)	1 review
	ACM International Conference on Mobile Human-Computer Interaction (2021)	1 review
Student Volunteer	*IEEE Conference on Virtual Reality and 3D User Interfaces* (VR)	2020, 2021
	IEEE International Symposium on Mixed & Augmented Reality (ISMAR)	2019
University of Maryland	GAMMA Lab Twitter account admin	2023 - 2024
	Graduate admissions application reviewer	2019 - 2024
	Girls Talk Math summer camp problem set reviewer	2021
	Graduate school application mentor	2020
Davidson College	Math & CS department student representative	2018 - 2019
	Davidson College ACM chapter co-founder	2018 - 2019

SKILLS

Computing Skills	C++, Python, C#, R, Unity3D, Unreal Engine, PsychoPy, D3.js, git, LaTeX, Windows, Linux
Research Areas	Virtual reality, applied visual perception, computer graphics, artificial intelligence for graphics and simulation

MEDIA COVERAGE

- *New Faculty at NYU Tandon CSE* - **NYU**
  Link: https://engineering.nyu.edu/news/new-faculty-fall-2024#Niall

- *This computer scientist is making virtual reality safer* - **Science News Explores**
  Link: https://www.snexplores.org/article/computer-scientist-safer-virtual-reality
- *Graduate Student Niall Williams Awarded Link Foundation Fellowship* - **UMD CS**
  Link: https://www.cs.umd.edu/article/2022/06/graduate-student-niall-williams-awarded...
- *This New Algorithm Lets You Explore Virtual Reality by Walking Naturally* - **UMIACS**
  Link: https://www.umiacs.umd.edu/about-us/news/new-algorithm-lets-you-explore...
- *Graduate Student Niall Williams Awarded Honorable Mention, Best Paper at 2022 IEEE VR* - **UMD CS**
  Link: https://www.cs.umd.edu/article/2022/03/graduate-student-niall-williams-awarded...